Identification of Disinformation types in LLMs

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Abstract

Over time, the field of LLMs has rapidly advanced, demonstrating not only its ability to produce coherent and contextually relevant responses across a wide range of tasks but also its equal potential to cause problems, including the spread of disinformation, factual inaccuracies, propagation of biases, and more. This research uses prompting techniques to detect various types of disinformation with the aid of dataset created by our own derived using subreddits. Our findings show that Few Shot prompting method performs well across various LLMs, such as Llama, Gemma, and Mistral, considering each label. When compared to other models, Llama exhibits a higher classification accuracy rate of 53%, while Mistral has the lowest accuracy rate of 38.5%. Regarding SLM model, accuracy remains low in comparison with LLMs. Contrary to the initial notion that satirical content would be challenging to classify, misleading content turns out to be tricky when compared to categories like truth, satire, misleading content, and imposter content.

Keywords

LLM, Classification, Llama, Mistral, Gemma, True, Satire, Imposter content, Misleading content, Disinformation, type, Binary, bias, position, prompt

1 Introduction

No wonder LLMs have proven to be powerful tools for handling complex language tasks. However, a critical concern in their widespread usage is the factuality of the information generated by them. These models are generally pre-trained on large datasets sourced from the internet containing a combination of opinions, accurate information, outdated facts, sarcasm, and fake news.

Though LLMs are built to mimic human interactions, they lack the inherent capability of understanding the verification mechanism [2]. As a result, the models end up generating text containing misinformation, which sometimes leads to false narratives. Ensuring the factual accuracy of the generated content becomes crucial in high-stake domains such as healthcare, law, and more, putting forth serious consequences. Hence, there is a need for post-generation verification mechanisms to lower the impact caused by the large language models.

While a number of studies have been carried out recently to categorize the data produced by LLMs as "true" or "false" or even to label false news into various categories. Few attempts have been made to compare the predictions of SLMs with LLMs by fine-tuning an SLM model, such as Roberta, for different dataset sizes and highlighting the intrinsic characteristics of LLMs, like positional, attention, and label bias.

In this study, we explore positional bias [9] through binary classification, label bias using prompting approaches such as zero shot, few shot, and COT, and how built-in transformers exhibit a feature known as attention bias [2]. To do this, we sourced our own dataset from Reddit [6]. The prompts tested across models such as Llama, Gemma, and Mistral remain consistent, and the accuracy to classify various types of disinformation such as true, misleading, satire, and imposter are recorded to determine the most efficient prompting method and the most deceptive disinformation category.

2 Related Works

In analyzing most available datasets, a common pattern emerges: they frequently employ broad classifications like "true" and "false." Although these datasets often contain millions of rows and are well-developed, they struggle to capture the nuanced nature of disinformation. However, some datasets, such as NELA-GT-2018 (Nørregaard et al., 2019) [1], LIAR (Wang, 2017) [10], and FakeNewsCorpus [7], offer more detailed classifications. While some datasets, like NELA-GT-2018 (Nørregaard et al., 2019) [1] , account for heterogeneous sources of information, others focus on specific areas, such as politics and gossip. Consequently, these datasets provide limited context due to their restricted sources and classification types.

Capabilities of LLMs to generate as well as generate disinformation have researched significantly in recent years. Studies such as Jiang, Bohan et al ., 2024 [4] and Leite, João A., et al. 2023 [5] explored how the LLMs perform under different prompting techniques to classify disinformation. These studies are instrumental in understanding the capabilities of LLMs to detect disinformation. However these studies focused on the broad classification of disinformation into true or false. Apart from that these analyses used datasets which focused on a specific domain like mentioned earlier.

Positional bias in LLMs refers to the tendency to rely on the positions of the tokens in the input, which can adversely affect them especially in the classification tasks. Vaswani et al., 2017 [9], was a path breaking study which introduced transformer architecture. Transformer based LLMs like GPT, Llama etc use positional encoding which leads to the positional bias. Yu, Yijiong, et al., 2024 [12] and Hsieh, Cheng-Yu, et al., 2024 [2] explained the tendency of LLMs to give priority for the tokens in the beginning and ending of the input.

Though LLMs are generally believed powerful, Hu, Beizhe, et al., 2024 [3] reveals that LLMs fails to outperform fine tuned small LM in different prompting techniques. Other than completely depending on LLMS to detect disinformation, the authors proposes a model in which LLMs act as advisor to the fine-tuned SLMs. Like the studies mentioned early, this analysis also classifies data into binary and focused on GossipCop (Shu et al., 2020) [8].

3 Methodology

3.1 Dataset

Most of the datasets available are built for usual binary classification rather than fine-grained classification.Our tailor made dataset focused on distinguishing each instance of news into 4 categories. We briefly describe each category:

- True: Indicates that the news is true.
- Satire/Parody: Indicates if content is twisted or misinterpreted in a satirical or humorous way.
- Misleading Content: Corresponds to the news in which the information is intentionally manipulated to fool audience.
- Imposter Content: Represents content generated by bots.

The studies by Wardle, C., & Derakhshan, H., 2017 [11] shows that there are 4 more categories of disinformations such as Fabricated Content, False Connection, False Context and Manipulated Content. These types are not considered as they rely on the multimodal data.

We sourced our data from Reddit, a social media which follows a thread and discussion model where users can post their submissions on different subreddits. It is also one of the top 20 websites in the world by traffic. As depicted in the Table 1 , we relied on these subreddits identified as sources in Fakeddit (Nakamura, K., Levy, S., & Wang, W. Y., 2019 [6] for each of the disinformation categories. Each subreddit has its own theme and set of rules managed by moderators. Active reddit users can upvote, downvote and submit comments on submissions.

Reddit submissions were gathered through the pullpush.io API, which is a tool used to index and retrieve content posted by Reddit users on the platform. We collected submission title, body text, and metadata such as score and upvote ratio. Ensuring the reliability of the data is essential. We have multiple stages of data processing. The initial one is offered via the subreddit pages. Every subreddit is moderated to make sure that posts are relevant to the theme of the subreddit. These moderators are responsible for deleting posts that break the rules. Refinement advances when data gathering takes place. Reddit's upvote/downvote system is employed during this phase. It is meant to show approval for a post and highlight posts that are not relevant or off-topic based on their score. Therefore, we eliminated any entries with a rating below 1 in order to guarantee the credibility of our data. It is assumed that improper or off-topic posts in a subreddit will either be deleted or receive a negative score of less than 1.

3.2 Zero Shot Prompting

Zero-shot prompting is made use to classify the given content into predefined categories without any prior training on specific examples. The model is provided with instructions to perform the classification task based solely on its existing knowledge. Below is an example prompt used in this methodology:

```
Prompt - 1 (Llama)
Content categories are as follows:
True
Satire
Imposter Content
Misleading Content
Analyze the type of content and return the
corresponding label.
```

```
Prompt - 2 (Llama)
Content categories are as follows:
Imposter Content
Satire
Misleading Content
True
Analyze the type of content and return the
corresponding label.
```

The model is expected to assess the given content and categorize it into one of the specified categories (True, Satire, Imposter Content, or Misleading Content).

3.3 Chain of Thoughts Prompting

Here, we use chain-of-thoughts prompting to not only classify the content but also explain the reasoning process. The model is guided to assess the given content systematically by considering multiple factors before arriving at a conclusion. Below is a general prompt used in this methodology:

```
General Prompt
Analyze the type of the content enclosed in square
brackets, and determine if it is true content, satire,
 misleading content, or imposter content. Explain
your reasoning step by step and then return the
answer as the corresponding content label "true
content" or "satire" or "misleading content" or "
imposter content".

   Step-by-step reasoning:
   1. Identify the primary purpose of the content (
   inform, entertain, deceive, etc.).
   2. Check for factual accuracy and sources.
   3. Determine if there is any exaggeration, humor,
    or irony.
   4. Check for any signs of manipulation or
   alteration.
   5. Determine if the content has been entirely
   fabricated.
```

The model is expected to follow these steps to provide a detailed analysis of the content before determining its label. This method

Table 1: List of subreddits used in Dataset

Subreddit	Category	URL
neutralnews	True	https://www.reddit.com/r/neutralnews
upliftingnews	True	https://www.reddit.com/r/upliftingnews
usnews	True	https://www.reddit.com/r/usnews
theonion	Satire	https://www.reddit.com/r/theonion
waterfordwhispersnews	Satire	https://www.reddit.com/r/waterfordwhispersnews
savedyouaclick	Misleading	https://www.reddit.com/r/savedyouaclick
subredditsimulator	Imposter	https://www.reddit.com/r/subredditsimulator
subsimulatorgpt2	Imposter	https://www.reddit.com/r/subsimulatorgpt2

allows for a more comprehensive evaluation, leveraging the model's ability to break down complex tasks into simpler steps.

3.4 Binary Classification

Binary classification was conducted for each class separately from the categories: True, Satire, Misleading Content, and Imposter Content. Each class was isolated and classified individually to examine whether focusing on individual categories could yield better accuracy compared to combined classification.

By performing binary classification, we also aimed to assess if LLMs exhibits positional bias. This approach helps to identify if the model favors certain content types when labels are presented in different orders or contexts.

3.5 24 Permutations

To investigate positional bias in the Large Language Model (LLM), we explored all possible positional arrangements of the four content classes: True, Satire, Misleading Content, and Imposter Content. This involved changing the order of the classes in all 4! (24) possible ways.

By classifying the content using these 24 different permutations, we aimed to determine if the LLM's performance varied based on the positional order of the categories, thereby revealing any potential biases in its responses.

3.6 SLM vs LLM

This approach compares SLM with LLM to evaluate their performance on the same dataset. RoBERTa was chosen as the SLM, and LlaMA, Gemma, and Mistral were employed as the LLMs.

Firstly, we used the dataset for both SLM and LLM models without any fine-tuning to assess their out-of-the-box capabilities. Post which we fine-tuned RoBERTa to examine if training could enhance its performance compared to the LLMs. This comparison allowed us to identify which model type is more effective under various conditions.

4 Results

In this section, we present the results obtained for each model. We report the recall, precision, and F1 scores obtained by all the models for various prompting techniques. Comparing the accuracy of various prompting technique helps us to find the best prompting approach. Moreover, we are also interested in knowing about the variety of bias exhibited by LLMs.

4.1 Prompting techniques

Among several prompting strategies, Zero shot, Few Shot and Chain-of-Thoughts are chosen considering the relevance to our study. The same prompts are used for recording the accuracy, recall, precision, and F1 scores in models such as Llama, Gemma, and Mistral.

Table 2: Llama - Comparison of Prompting Techniques

Prompts	Accuracy	Precision	Recall	F1-Score
Zero Shot	0.27	0.21	0.27	0.19
CoT	0.4	0.39	0.39	0.32
Few Shots	**0.53**	**0.68**	**0.54**	**0.54**

Table 3: Gemma - Comparison of Prompting Techniques

Prompts	Accuracy	Precision	Recall	F1-Score
Zero Shot	0.46	0.53	0.46	0.42
CoT	**0.71**	**0.71**	**0.71**	**0.71**
Few Shots	X	X	X	X

It is evident that with the Few Shot prompting technique, Llama produces the highest accuracy of 53% and F1 score of 0.54, while the other methods end up with zero or lesser values.

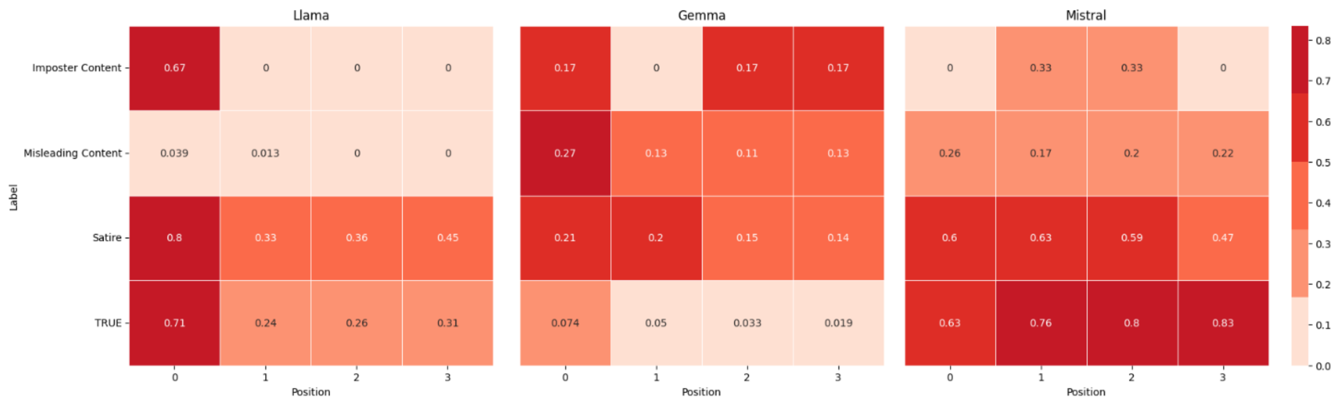

Figure 1: Heat Map - 24 Permutations of Llama, Gemma and Mistral

Table 4: Mistral - Comparison of Prompting Techniques

Prompts	Accuracy	Precision	Recall	F1-Score
Zero Shot	**0.4**	0.38	**0.4**	**0.37**
CoT	0.23	0.50	0.22	0.28
Few Shots	0.38	**0.51**	0.38	0.32

4.2 Positional Bias

While experimenting with Zero Shot prompting technique by placing labels at different positions, it is noticeable that not all positions received similar weightage. This is due to the fact that LLMs are built using sequence-based architecture which assign disproportionate weights to certain tokens based on their sequence in the input.

The heat maps shown above in Figure 1 demonstrates how the accuracy vary with respect to position when labels are called in sequence using Zero Shot prompting technique.

Interestingly, Mistral outperforms all the other models by showcasing less positional bias.

4.3 Results of Binary Classification

In order to cross-validate the previous observation on positional bias, we ran individual labels to record the accuracy and compare it with the prompt results when called in a sequence.

Table 5: Accuracy of Models on Binary Classification

Prompts	Llama	Gemma	Mistral
True	0.907	0.458	0.85
Satire	0.81	0.2	0.84
Misleading	0.957	0.28	0.72
Imposter	0	0	0

Table 6: F1-Score of Models on Binary Classification

Prompts	Llama	Gemma	Mistral
True	0.72	0.3	0.54
Satire	0.47	0.2	0.52
Misleading	0.48	0.21	0.41
Imposter	0	0	0

As expected, the accuracy is relatively higher when called individually than in sequence (Table 2, 3 and 4) which confirms the existence of positional bias. However a pattern can be observed .

4.4 Attention Bias

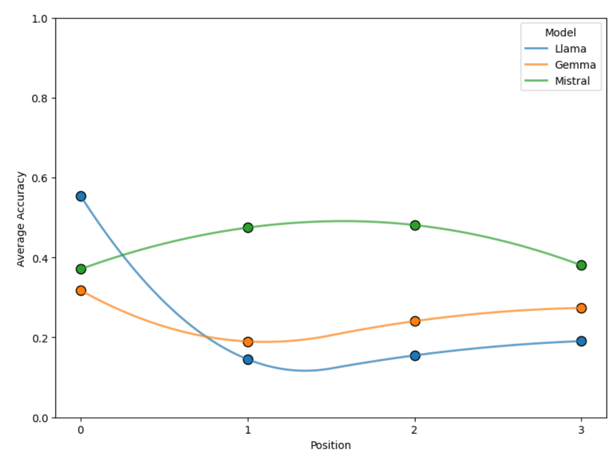

Figure 2: Attention Bias

Talking about the pattern observed in the previous section, the phenomena is called Attention bias where either the beginning and end or the middle of the sequence gets more emphasis.

The graph above in Figure 2 leaves us with two shapes. Llama and Gemma depicts a U-shaped curve indicating the importance given to start and end of the sequence whereas Mistral presents a inverted U-shaped curve pointing the attention given to middle of the sequence.

In case of Llama and Gemma , the attention bias is called as U-shaped bias. On the other hand, the attention bias in Mistral is known as Found-in-the-middle.

4.5 SLM v/s LLM

Fine-tuned Roberta is ran across varying dataset sizes to record the efficiency of how well it can classify the labels in comparison with LLMs.

The graph here shows the performance of Roberta with 500 and 1000 rows of dataset. It managed to provide accuracy of 23% with the dataset of 500 rows and 30% with the dataset of 1000 rows. Surprisingly a spike in accuracy is seen as the dataset size increases.

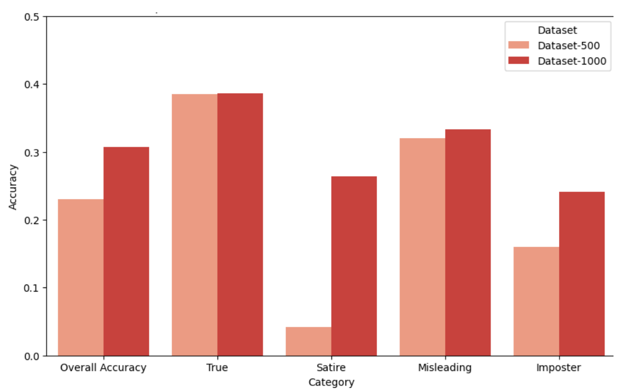

Figure 3: Comparison on 500 v/s 1000 dataset rows

5 Discussion

Focusing on the efficient prompting technique and the most deceptive disinformation category. Llama records a accuracy of 53% with Few Shot. As seen in the Tables 2, 3 and 4, Few Shot not only gave the highest accuracy but also considered all the positions while labeling. Altogether making it the most efficient prompting technique.

Table 7: Llama - Few Shots

Prompts	Accuracy	Precision	Recall	F1-Score
True	0.389	0.67	0.39	0.49
Satire	0.769	0.69	0.77	0.73
Misleading	**0.121**	0.71	0.12	**0.21**
Imposter	0.879	0.68	0.88	0.76

Table 8: Mistral - Few Shots

Prompts	Accuracy	Precision	Recall	F1-Score
True	0.131	0.78	0.13	0.22
Satire	0.93	0.32	0.93	0.47
Misleading	**0.032**	0.25	0.03	**0.06**
Imposter	0.44	0.7	0.44	0.54

Considering the F1-scores and accuracy of each label, Misleading content has 0.12 accuracy and F1 score of 0.21 for Llama (Table 7), 0.03 accuracy and F1 score of 0.06 for Mistral (Table 8). Thus by looking at the low F1 scores, we can say that the content does not cover all information needed by LLM to categorise it as Misleading and making it the most deceptive disinformation category.

6 Conclusion

The spread of fake news is not new, but its reach and impact have been amplified with advancements in the domain of LLMs. As a result, there is a need to identify the type of misinformation to validate the factuality of information generated by LLMs.

Apart from recent works, our research answers the importance of prompting strategies in classification and how the sophisticated mimicry of factuality of misleadling content turned it to be the trickiest to classify.

Additionally, LLM outperforms SLMs due to the accuracy's dependence on the size of the dataset used to train them.

In future work, we aim to pre-train the SLM model with a larger dataset if given the necessary GPU capacities in order to uncover hidden capabilities and explore Gemma's responses to COT and Few Shot prompting strategies

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7 Abbreviations

Table 9: The following abbreviations are used

Abbreviation	Definition
LLM	Large Language Model
SLM	Small Language Model
COT	Chain of Thoughts

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